

Patient Information

Age

44 - +

Sex

1 ▾

Chest Pain Type

1 ▾

Resting Blood Pressure

150 - +

Cholesterol

200 - +

Fasting Blood Sugar >120

0 ▾

Rest ECG

0 ▾

Maximum Heart Rate

150 - +

Exercise Induced Angina

0 ▾

Oldpeak

1.00 - +

Slope

0 ▾

Major Vessels

0 ▾

Thal

0 ▾

♥ Heart Disease Prediction with Explainable AI

Predict

Prediction Result

⚠ Heart Disease Risk Detected

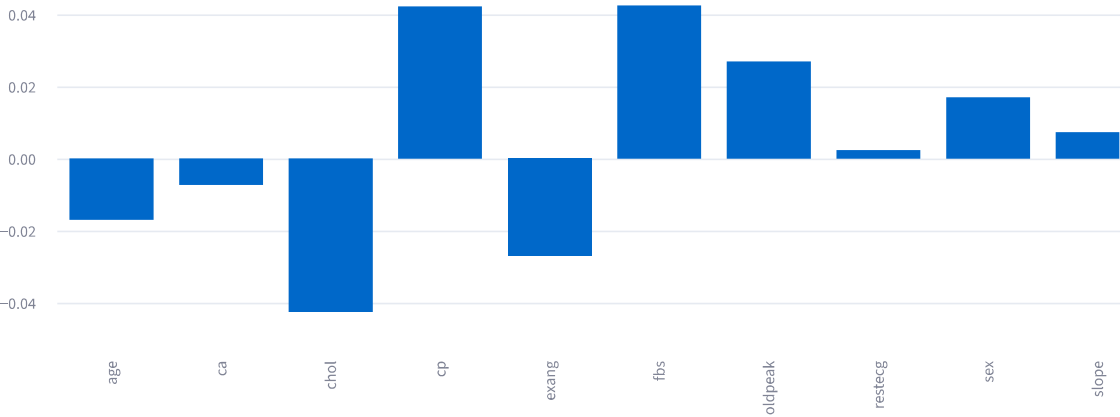
Probability: 72.00%

Heart Disease Probability

72.00%

SHAP Explainability

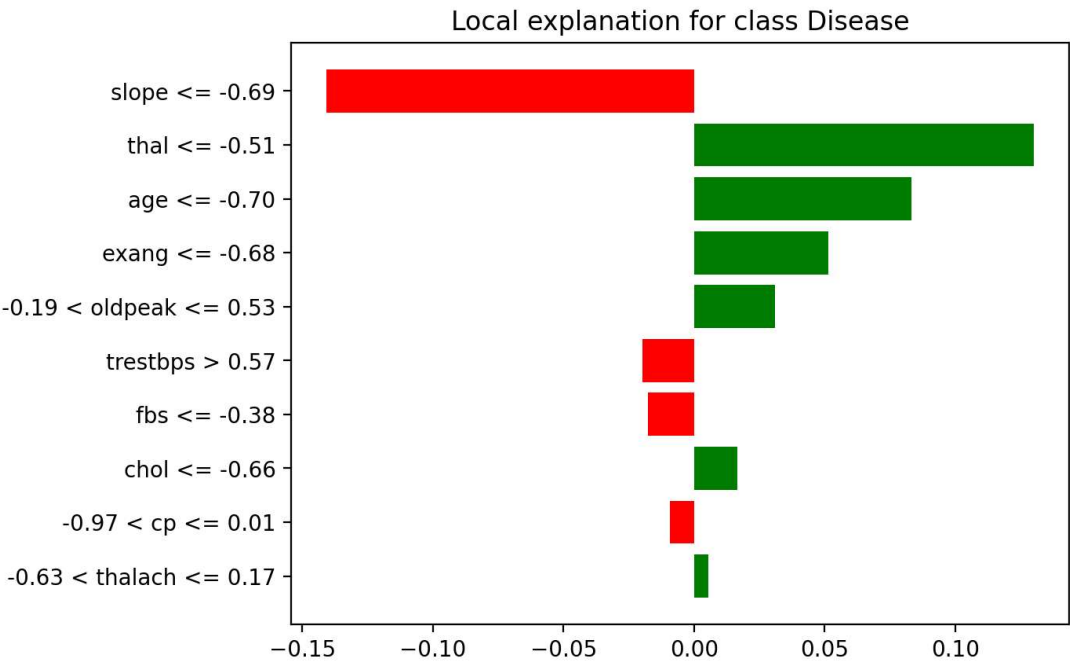
Feature Importance



Top Risk Factors

- fbs increases risk
- chol decreases risk
- cp increases risk
- trestbps decreases risk
- exang decreases risk

LIME Explanation



Clinician-Friendly Explanation

- slope ≤ -0.69 reduces the likelihood of heart disease.
- thal ≤ -0.51 increases the likelihood of heart disease.
- age ≤ -0.70 increases the likelihood of heart disease.
- exang ≤ -0.68 increases the likelihood of heart disease.
- $-0.19 < \text{oldpeak} \leq 0.53$ increases the likelihood of heart disease.
- trestbps > 0.57 reduces the likelihood of heart disease.
- fbs ≤ -0.38 reduces the likelihood of heart disease.
- chol ≤ -0.66 increases the likelihood of heart disease.
- $-0.97 < \text{cp} \leq 0.01$ reduces the likelihood of heart disease.
- $-0.63 < \text{thalach} \leq 0.17$ increases the likelihood of heart disease.

Educational use only. Not a medical diagnosis tool.